Chapter 6 Practice

Sheena Bennett

```
library(cancensus)
```

```
Warning: package 'cancensus' was built under R version 4.4.3
```

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'readr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.5.1
v ggplot2   4.0.3     v tibble    3.2.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
get_census(
  dataset = "CA1996",
  level = "Regions",
  vectors = "v_CA16_1",
    regions = list(PR = c("35"))
  )
```

```
Reading vectors data from local cache.
```

```
# A tibble: 1 x 9
  GeoUID Type  `Region Name` `Area (sq km)` Population Dwellings Households
  <chr>  <fct> <fct>                  <dbl>      <dbl>     <dbl>      <dbl>
1 35     PR    Ontario               987898.   10753573   3951326    3924515
# i 2 more variables: C_UID <chr>, `v_CA16_1: Age Stats` <dbl>
```

```
get_census(
  dataset = "CA1996",
  level = "Regions",
  vectors = "v_CA16_1",
  regions = list(PR = c("35"))
)
```

```
Reading vectors data from local cache.
```

```
# A tibble: 1 x 9
  GeoUID Type  `Region Name` `Area (sq km)` Population Dwellings Households
  <chr>  <fct> <fct>                  <dbl>      <dbl>     <dbl>      <dbl>
1 35     PR    Ontario               987898.   10753573   3951326    3924515
# i 2 more variables: C_UID <chr>, `v_CA16_1: Age Stats` <dbl>
```

```
data_1996 <-
  get_census(
    dataset = "CA1996",
    level = "Regions",
    vectors = "v_CA1996_2",
    regions = list(PR = c("35"))
  )
```

```
Reading vectors data from local cache.
```

```
data_1996_BC <-
  get_census(
    dataset = "CA1996",
    level = "Regions",
    vectors = " v_CA1996_2",
    regions = list(PR = c("59"))
  )
```

```
Reading vectors data from local cache.
```

```
data_1996_BC$Population
```

```
[1] 3724500
```

```
data_1996$Population
```

```
[1] 10753573
```

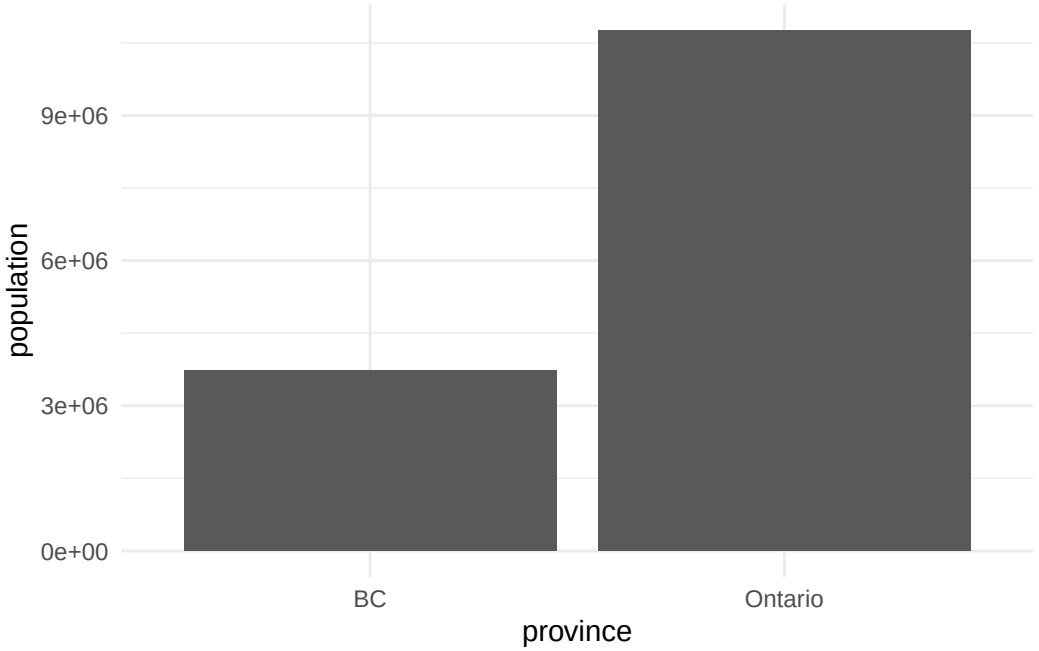
Comparing Population in British Columbia and Ontario

In this section, I compare the 1996 population of British Columbia and Ontario using Canadian census data retrieved with the `cancensus` package. I first create a table showing the population of each province and then visualize the comparison using a bar graph.

```
BC_ON_Data <- data.frame(
  id = c(1, 2),
  name = c("Ontario" , "BC"),
  population = c(data_1996$Population, data_1996_BC$Population)
)
BC_ON_Data
```

```
  id    name population
1  1 Ontario   10753573
2  2      BC    3724500
```

```
BC_ON_Data |>
  ggplot(aes(x = name, y = population)) +
  geom_col() +
  theme_minimal() +
  labs(x = "province", y = "population")
```


9e+06
6e+06
3e+06
0e+00
population
BC
Ontario
province